

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Application-Based Questions

Weightage: 40%

QUESTIONS: Total Marks: 30

1: Online Symposium Portal

A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:

```
<form action="/register" method="POST">
  <input type="text" name="name">
  <input type="email" name="email">
  <button>Submit</button>
</form>
```

When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly.

Questions:

1. Identify appropriate **HTML5 semantic elements** missing in the structure.

A.

Missing HTML5 semantic elements:

1. `<header>` – Displays the page title or logo.
2. `<main>` – Contains the main content.
3. `<section>` – Groups the registration form.
4. `<h1>` – Provides a heading for the page.
5. `<label>` – Associates labels with input fields.
6. `<footer>` – Displays copyright or contact information

2. Modify the structure to make it **standards-compliant**.

A.`<!DOCTYPE html>`

`<html lang="en">`

`<head>`

`<meta charset="UTF-8">`

`<meta name="viewport" content="width=device-width, initial-scale=1.0">`

`<title>Symposium Registration</title>`

`</head>`

`<body>`

`<header>`

`<h1>Online Symposium Registration</h1>`

`</header>`

`<main>`

`<section>`

`<form action="/register" method="POST">`

```
<label for="name">Name:</label>

<input type="text" id="name" name="name" required><br><br>

<label for="email">Email:</label>

<input type="email" id="email" name="email" required><br><br>

<button type="submit">Submit</button>

</form>

</section>

</main>

<footer>

    <p>&copy; 2026 College Symposium</p>

</footer>

</body>

</html>
```

3. Interpret the **HTTP request generated** during form submission.

A. When the student clicks Submit, the browser sends an HTTP POST request.

Example:

```
POST /register HTTP/1.1
Host: www.college.edu
Content-Type: application/x-www-form-urlencoded
```

```
name=Ganesh&email=ganesh@example.com
```

4. Analyze the **HTTP response cycle** from server to browser.

A. Student fills the form.

1. Browser sends an HTTP **POST** request.
2. Server validates the submitted data.

3. Server stores the registration details in the database.
4. Server sends an HTTP response.

Example Response:

HTTP/1.1 200 OK
Content-Type: text/html

Registration Successful!

6. Browser displays the confirmation page to the student.
5. Suggest improvements for **usability and accessibility**.
- A.** Use `<label>` for every input field.
Add the `required` attribute to mandatory fields.
Use appropriate input types such as `email` and `tel`.
Display clear error messages for invalid input.
Make the form keyboard accessible.
Use semantic HTML elements (`header`, `main`, `section`, `footer`).
Ensure good color contrast for readability.
Add placeholders or helpful instructions where needed.

2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:

```
<html>
  <head>
    <title>Company</title>
  </head>
  <body>
    <div>
      <h1>Welcome
      <p>About us
    </div>
  </body>
</html>
```

Questions:

1. Identify **improper nesting and missing closing tags**.

A. Errors in the given HTML:

- `<h1>` tag is not closed.

- `<p>` tag is not closed.
- Missing `<!DOCTYPE html>`.
- Missing `<meta charset="UTF-8">`.

2. Analyze how different browsers may interpret this structure.

A. Some browsers automatically close the missing tags and display the page correctly. Other browsers may render the content differently, causing layout inconsistencies. Older browsers may not correctly interpret the incomplete HTML structure. This results in cross-browser compatibility issues.

3. Identify **missing semantic elements** (e.g., header, section).

A. The following semantic elements are missing:

- `<header>` – For the page heading.
- `<main>` – For the main content.
- `<section>` – To group related content.
- `<footer>` – For footer information (optional).

4. Provide a **corrected W3C-compliant HTML structure**.

A, `<!DOCTYPE html>`
`<html lang="en">`
`<head>`
`<meta charset="UTF-8">`
`<meta name="viewport" content="width=device-width, initial-scale=1.0">`
`<title>Company</title>`
`</head>`

`<body>`

`<header>`
`<h1>Welcome</h1>`
`</header>`

`<main>`
`<section>`
`<p>About us</p>`
`</section>`
`</main>`

`</body>`
`</html>`

5. Suggest techniques to ensure **cross-browser compatibility**.

A.

1. Use valid W3C-compliant HTML.
2. Always close all HTML tags properly.
3. Use semantic HTML elements.
4. Test the website in multiple browsers (Chrome, Firefox, Edge, Safari).
5. Use CSS Reset or Normalize CSS.
6. Include `<!DOCTYPE html>` at the beginning of every HTML page.
7. Validate the HTML and CSS using W3C validators.

3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

```
<form action="/submit" method="GET">
  <input type="text" name="username">
  <input type="password" name="password">
  <button type="button">Login</button>
</form>
```

Questions:

1. Identify issues in the **form configuration**.

A.

1. The button type is `button` instead of `submit`.
2. Using the **GET** method for sending a password is insecure.
3. Input fields do not have `<label>` elements.
4. `required` validation is missing.

2. Analyze why the **HTTP request is not triggered**.

A.

The button is defined as:

```
<button type="button">Login</button>
```

A button with `type="button"` performs no default action.

Therefore, clicking the button does not submit the form, so no HTTP request is sent. It should be:

```
<button type="submit">Login</button>
```

3. Interpret the role of **GET method in this scenario**.

A. The **GET** method sends form data in the URL.

Example:

```
/submit?username=Ganesh&password=12345
```

Since passwords appear in the URL, GET is **not secure** for login forms.

POST should be used to send sensitive information.

4. Propose a **corrected implementation**.

A.

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Login Form</title>
```

```
</head>
```

```
<body>
```

```
<form action="/submit" method="POST">
```

```
  <label for="username">Username:</label>
```

```
  <input type="text" id="username" name="username" required><br><br>
```

```
  <label for="password">Password:</label>
```

```
  <input type="password" id="password" name="password" required><br><br>
```

```
  <button type="submit">Login</button>
```

```
</form>
```

```
</body>
```

```
</html>
```

5. Suggest improvements for **secure data transmission**.

A.

1. Use the **POST** method instead of GET.
2. Use **HTTPS** to encrypt data during transmission.
3. Validate user input on both client and server.
4. Use strong password hashing (e.g., bcrypt) on the server.
5. Enable CSRF protection for forms.
6. Avoid displaying sensitive information in URLs.

Code of Conduct:

I Abburi Ganesh Ram Kamal (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand